Decentralized Deep Reinforcement Learning for Cooperative Multi-Agent Flight Trajectory Planning in Adverse Weather

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ABSTRACT

Climate change has led to an increased frequency of adverse weather events, particularly thunderstorms, which poses significant safety and efficiency challenges for en route aircraft operations. These conditions require multi-agent cooperative trajectory planning to avoid both dynamic thunderstorms and other aircraft. Current approaches often rely on centralized or single-agent methods, lacking the coordination and robustness needed for scalable solutions in response to fast-evolving aircraft or thunderstorms paths. This paper introduces a novel multi-agent cooperative framework for autonomous trajectory planning, where each aircraft operates as an autonomous agent, interacting with the dynamic airspace and weather environment to learn a shared policy for coordinated decision making. The problem is modeled as a Decentralized Markov Decision Process (DEC-MDP) and solved using an Independent Deep Deterministic Policy Gradient (IDDPG) algorithm, which integrates shared neural networks to improve training efficiency and handle unexpected traffic diversions. Simulation results, including real-world test cases, demonstrate the framework's robustness, scalability, and generalization across diverse traffic and weather conditions. The proposed IDDPG algorithm consistently outperforms a multi agent-based Fast Marching Tree algorithm in maintaining aircraft separation and optimizing rerouting efficiency, contributing to both the air traffic management and autonomous multi-agent system communities by advancing decentralized deep reinforcement learning in dynamic, safety-critical contexts.

KEYWORDS

Air traffic management, multi-agent systems, deep reinforcement learning, shared neural network, cooperative path planning

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1 INTRODUCTION AND BACKGROUND

Climate change is impacting air transport through adverse weather, particularly thunderstorms, with a rise in the frequency and severity [9]. These weather phenomena are highly disruptive, causing aircraft to lose separation and leading to severe turbulence, sometimes resulting in severe injuries [4]. The adverse weather has also, for the first time, become the leading reason of en-route air traffic flow management delays in Europe [9]. These dynamic thunderstorm cells can develop rapidly, obstructing nominal flight paths and requiring immediate and effective trajectory planning [3]. The challenge intensifies when multiple aircraft are involved, each needing to make real-time adjustments to avoid both other aircraft and evolving thunderstorm cells [15]. This is especially critical in oceanic airspace, where air traffic control coordination is limited or unavailable. In such conditions, the absence of proper coordination among multiple aircraft can lead to numerous loss-of-separation incidents [36]. Addressing this issue is crucial, given the increasing unpredictability and intensity of weather phenomena, which highlights the need for automated, cooperative tools to ensure safety and efficiency in congested airspace.

Several automation tools have been developed to assist pilots and ATCOs in managing complex tasks like conflict resolution and flight rerouting [11, 12, 27]. However, existing research still has several limitations. Current models often treat the paths of surrounding flights as static obstacles [42], leading to recalculations whenever an aircraft or thunderstorm changes course. This not only reduces robustness but also increases computational burdens, making it difficult to handle unpredictable environments effectively. Additionally, existing methods largely rely on the First-Come-First-Served principle [19], generating conflict-free trajectories for individual aircraft without enabling cooperative decision-making. This limits the ability to manage multiple aircraft collaboratively, reducing overall safety and efficiency. Moreover, the fast-changing nature of thunderstorm conditions and the diverse combinations of traffic flow and weather patterns require a scalable and computationally efficient framework, which current models fail to provide.

These limitations motivate us to explore a cooperative, robust, and scalable multi-agent framework for multi-aircraft trajectory planning under dynamic conditions. The proposed framework aims to: (a) effectively handle trajectory planning in dynamic and complex airspace environments; (b) maintain robustness against trajectory dependencies, accommodating disruptions without recalculating all trajectories; and (c) generalize effectively to unseen combinations of traffic and weather patterns. In summary, this paper contributes to decentralized deep reinforcement learning for innovative applications of multi-agent systems in safety-critical ATM fields, with the following aspects:

- We propose a multi-agent framework using decentralized reinforcement learning for multi-aircraft cooperative trajectory planning (MACTP) under rapidly evolving thunderstorms. This approach addresses challenges related to trajectory dependency and robustness in the face of unpredictable airspace disruptions, such as emerging thunderstorm cells.
- We develop a decentralized Markov Decision Process (DEC-MDP) model for cooperative planning, solved using an Independent

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Deep Deterministic Policy Gradient (IDDPG) algorithm. By integrating shared neural networks, agents exchange state and policy information during training and leverage each other's experiences, leading to improved convergence rates and more coordinated decision-making.
- We evaluated the proposed framework on complex multi-aircraft planning tasks across diverse traffic and adverse weather scenarios, where our approach outperformed the multi-agent baseline algorithm in maintaining safe separation and improving robustness of aircraft operations.

2 RELATED WORKS

The multi-aircraft cooperative trajectory planning (MACTP) problem is related to trajectory planning and conflict resolution in air traffic management [28]. This section reviews existing literature on these topics, focusing on traditional methods, general multi-agent pathfinding, and reinforcement learning-based approaches.

2.1 Traditional Approaches for Aircraft Trajectory Planning

Traditional trajectory planning under thunderstorm conditions has focused on both static and dynamic cells, utilizing geometry, optimization, and heuristics. Erzberger et al. [10] pioneered automated conflict resolution tools, highlighting the need for decentralized, real-time solutions. Ng et al. [25] and Kamgarpour et al. [21] tackled rerouting under weather uncertainties, using dynamic programming and constrained optimization. Gauthier's work [33] studied a multi-agent decentralized approach for unmanned ATM, tackling conflict resolution using distributed optimization. While these works demonstrated benefits over centralized approaches, challenges remained in scalability and computational efficiency.

Other approaches, like those by Zhang et al. [41] and Hentzen [19] focused on the effects of uncertainties, but their methods were either time-consuming or lacked real-time applicability. Taylor et al. [38] addressed constraint avoidance with multi-objective optimization but struggled with real-time coordination in dynamic conditions. Other studies, such as Pannequin et al. [29] and Summers et al. [37], applied nonlinear and stochastic control to optimize aircraft trajectories but faced scalability and oversimplification issues. González-Arribas et al. [14] and Seenivasan et al. [34] tried to enhance robustness against thunderstorms using optimal control but struggled with scalability and coordination. Heuristic and sampling-based methods have also been explored. Andres et al. [2] applied RRT* for near real-time planning, while Guitart et al. [16] proposed sample-based path planning. Despite their effectiveness, issues of trajectory dependency and scalability persist.

In summary, current methods often assume static weather and focus on centralized or single-agent control, limiting coordination, scalability, and efficiency in dynamic, high-density environments.

2.2 Multi-Agent Path Finding Approaches

Multi-Agent Path Finding (MAPF) has been a focus in multi-agent research, aiming to plan collision-free paths for multiple agents [1]. Liu et al. [23] addressed task assignment and pathfinding in the multi-agent pickup and delivery (MAPD) problem, proposing offline algorithms to solve deadlock issues. Zhang et al. [40] added precedence constraints to improve real-world applicability. To make MAPF models more explainable, Shaull et al. [1] proposed a visualization-based explanation scheme for decomposing paths. While these methods are valuable, they often focus on 2D pathfinding, lacking applicability for 3D, such as aircraft path planning. Further advancements include collaborative deep reinforcement learning frameworks by Wu et al. [39], which improve Pareto solutions in multi-objective problems, and a machine learning-based approach by Huang et al. [20] to enhance computational efficiency. Luo et al. [24] proposed AlphaRoute, a reinforcement learning-based model for road traffic that achieves efficient route planning, while Alexey et al. [35] introduced decentralized Monte Carlo Tree Search for MAPF with limited communication.

Existing MAPF approaches focus primarily on ground robot navigation, which face challenges when applied to air traffic management under extreme weather. These methods could struggle with maintaining safe separation and adapting to the complex combination of dynamic thunderstorm cells and diverse traffic patterns.

2.3 Multi-Agent Reinforcement Learning for Trajectory Planning

Reinforcement learning (RL) has been applied to multi-agent systems for aircraft trajectory planning, with a focus on conflict detection and resolution (CDR). Early research explored RL-based multi-agent approaches in air traffic management and Urban Air Mobility [7]. Brittain and Wei [5, 6] developed a deep MARL model using Proximal Policy Optimization (PPO) for tactical CDR, showing scalability and efficiency in simulations. Pham et al. [32] introduced a two-layer Deep Deterministic Policy Gradient (DDPG) model to optimize aircraft vectoring for CDR under uncertainty, effectively managing high-density traffic. Zhao and Liu [43] integrated physics-based knowledge into RL, creating explainable CDR solutions, though challenges remained in practical air traffic control (ATC) applications. Chen et al. [8] studied RL model generalization, highlighting a trade-off between flight efficiency and separation as traffic density increased. Papadopoulos et al. [30] validated MARL models in simulated real-world settings but noted the need for alignment with ATC procedures.

Most RL applications in aircraft trajectory planning focus on isolated tasks like conflict resolution using centralized single-agent control. However, multi-aircraft trajectory planning under evolving thunderstorms requires cooperative, multi-agent approaches capable of addressing tasks like separation assurance, storm cell avoidance, and waypoint adherence simultaneously. This motivates a decentralized and cooperative approach in this study.

3 PROBLEM FORMULATION

3.1 Problem Statement

Rapidly developing thunderstorms pose significant challenges in busy airspace due to their unpredictability and swift onset. As shown in Figure 1, emerging thunderstorm cells can obstruct key air routes (e.g., L101 to L104), forcing multiple aircraft to deviate from planned paths. Managing these rerouted trajectories becomes particularly complex in high-density air traffic environments, especially under unstable thunderstorm conditions [15, 26].

A key concern is maintaining safe separation while minimizing additional flight distance. For instance, rerouted aircraft from L102 may conflict with others from L104, creating collision risk hotspots. Similarly, flights avoiding thunderstorms on L103 may intersect alternate routes, complicating traffic management and increasing collision risks.

Rerouted aircraft must avoid not only each other but also evolving storm cells. Despite deviations, flights must eventually rejoin nominal routes at designated exit waypoints ($w1, w2, w3, w4$) to continue toward their destinations. These potential conflicts highlight the need for a robust system capable of dynamically adapting to rapidly changing weather conditions and ensuring safe separation.

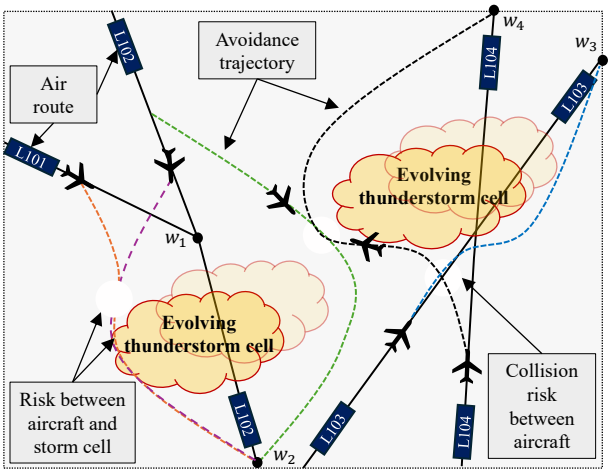


Figure 1: Multi-agent cooperative trajectory planning in dynamic airspace with evolving thunderstorm cells. Multiple aircraft must continuously adjust their routes to navigate around dynamic weather disturbances and maintain safe separation, highlighting the crucial need for coordination.

3.2 Mathematical Expressions

We develop a mathematical formulation with defined parameters, objectives, and constraints, which provides a clear scope and definition of the research problem.

Notations:

Aircraft: let $A = \{a_1, a_2, \ldots, a_n\}$ be the set of aircraft in the weather-affected airspace.

Exit waypoints: let $W = \{w_1, w_2, \ldots, w_l\}$ be the set of positions of exit waypoints.

Time: let $T = \{t_0, t_1, \ldots, t_{\text{final}}\}$ denote discrete time steps, where t_0 is the rerouting start time, and t_{final} is when an aircraft reaches its exit waypoint w_l and leaves the affected airspace.

Thunderstorm cells: let $O^W(t) = \{o_1^W(t), o_2^W(t), \ldots, o_j^W(t)\}$ be the set of thunderstorm cells' range at time t. Each cell acts as a time-dependent obstacle.

Centroid of the thunderstorm cell: $p_{O^W}(t)$ is the centroid position of thunderstorm cell O^W at time t.

Aircraft position: the two-dimensional position of aircraft a_i at time t is denoted by $p_i(t) \in \mathbb{R}^2$.

Safe separation: let $d_{\min}^{\text{flight}}$ be the separation minima between aircraft, and $d_{\min}^W$ be the minimum distance to thunderstorm cells.

Decision variables:

- $\Delta h_i(t)$: represents heading change value for aircraft i at t. As we assume fully autonomous operations, the heading change is treated as a continuous variable.
- $\alpha_i(t)$: an artificial variable denoting the heading of aircraft i at t.
- $p_i(t)$: an artificial variable denoting the position of aircraft i at t.

Objective functions:

One objective is to minimize the total distance each aircraft agent travels from its current position $p_i(t)$ at each time step to its respective exit waypoint $p_i(t_{\text{final}})$, minimizing the distance between consecutive positions. This approach encourages direct paths for aircraft, which may lead to deviations from their original routes if those routes were not the most efficient. Consequently, this model is particularly suited for airspace operations where avoidance of dynamic thunderstorm cells is required, but it may need modifications for non-weather airspace operations where more structured route adherence is necessary.

$$\min \sum_{i=1}^{n} \sum_{t=t_0}^{t_{\text{final}}} \|p_i(t) - p_i(t_{\text{final}})\| \tag{1}$$

Another objective is to minimize heading change maneuvers, as they increase aircraft agent interactions and airspace complexity, potentially leading to passenger injuries. We define Δh as a constant threshold for each heading change, making the term $\left|\frac{\Delta h_i(t)}{\Delta h}\right|$ in the range of $[0, 1]$.

$$\min \sum_{i=1}^{n} \sum_{t=t_0}^{t_{\text{final}}} \left|\frac{\Delta h_i(t)}{\Delta h}\right| \tag{2}$$

Constraints:

- Aircraft separation constraint: The distance between any two aircraft agent a_i and a_j at any given time t must be greater than or equal to the minimum separation distance $d_{\min}^{\text{flight}}$.

$$\|p_i(t) - p_j(t)\| \geq d_{\min}^{\text{flight}}, \quad \forall i, j \in \{1, \ldots, n\}, i \neq j, \forall t \in T \tag{3}$$

- Thunderstorm cell separation constraint: The distance of an aircraft (position $p_i(t)$) from the centroid of the thunderstorm cells ($p_{O^W}(t)$) at any given time t is greater than the minimum separation distance $d_{\min}^W$ plus the radius of the major axis of the thunderstorm cell ellipse $R_{\text{major_axis}}^W$.

$$\|p_i(t) - p_{O^W}(t)\| \geq d_{\min}^W + R_{\text{major_axis}}^W, \quad \forall i \in \{1, \ldots, n\}, \forall t \in T \tag{4}$$

- Exit waypoint reachability constraint: To ensure rerouted flights resume their nominal air routes, each aircraft must reach its assigned exit waypoint.

$$p_i(t_{\text{final}}) = w_{a_i}, \quad \forall i \in \{1, \ldots, n\} \tag{5}$$

- Aircraft dynamics: The position of each aircraft at each time step is updated based on its current position, velocity, and heading change. Let $p_i(t) = (x_i(t), y_i(t))$ represent the position of aircraft

i at t, $v_i(t)$ be the velocity, and $\Delta h_i(t)$ the heading change value. The position update model is as follows:

$$p_i(t+1) = \begin{bmatrix} x_i(t) \\ y_i(t) \end{bmatrix} + v_i(t)\Delta t \begin{bmatrix} \cos(\alpha_i(t+1)) \\ \sin(\alpha_i(t+1)) \end{bmatrix} \tag{6}$$

$$\alpha_i(t+1) = \alpha_i(t) + \Delta h_i(t) \tag{7}$$

where $\alpha_i(t+1)$ is the new heading angle after applying the heading change, $p_i(t+1)$ is the new position, and Δt is the time step duration.

The problem formulation is a multi-objective nonlinear and non-convex programming problem, which poses severe computational challenges. Therefore, it is impractical for time-sensitive applications like those encountered in dynamic air traffic management that require a solution almost in real-time. Additionally, certain adaptive and experience-based decision-making features, such as learning from past trajectory adjustments in similar weather conditions, cannot be effectively captured through mathematical formulations alone. These features rely on an agent's ability to accumulate experience over time, improving its decision-making through reinforcement learning in dynamic and uncertain environments. These need for handling multiple objectives, ensuring fast computation, and modeling implicit features render conventional model-based optimization and meta-heuristic algorithms insufficient, motivating us to explore learning-based methods to solve this problem.

4 DEC-MDP MODEL AND IDDPG SOLUTION

4.1 Decentralized Markov Decision Process

We reformulate the optimization problem as a Markov Decision Process (MDP) to address sequential decision-making in dynamic and uncertain environments. The complexity lies in the continuous adaptation required by each aircraft, making decentralized control suitable. In this setting, each aircraft acts as an independent agent, with its observations and state transitions, while all agents collectively aim to maximize a joint reward for the system, leading to the use of a decentralized MDP (DEC-MDP).

For a system of n agents, the DEC-MDP is defined by the tuple (S, A, R, P), where:

- $S = S_1 \times S_2 \times \cdots \times S_n$: Represents the joint state space, where S_i is the state space for agent i.
- $A = A_1 \times A_2 \times \cdots \times A_n$: Represents the joint action space, with A_i as the action space for agent i.
- $R(s_1, s_2, \ldots, s_n, a_1, a_2, \ldots, a_n)$: Represents the joint reward function for state-action pairs.
- $P = (P_1, P_2, \ldots, P_n)$: Represents the state transition probabilities, with each P_i as a function mapping from state-action pairs to rewards, denoted as $S_i \times S_i \times A_i \to R$.

The transition probabilities for joint states to $(s'_1, s'_2, \ldots, s'_n)$ given a joint action $(a_1, a_2, \ldots, a_n)$ are computed as:

$$P(s'_1, s'_2, \ldots, s'_n \mid s_1, s_2, \ldots, s_n, a_1, a_2, \ldots, a_n) = \prod_{i=1}^{n} P_i(s'_i \mid s_i, a_i) \tag{8}$$

This formulation is scalable to an increased number of agents, making it suitable for multi-aircraft trajectory planning, where each aircraft acts independently but with a shared objective. The DEC-MDP framework models dynamic interactions, stochastic weather, and the need for real-time decision-making. We detail each element of the DEC-MDP model below.

4.1.1 State Space. The state space s_t^i for aircraft i at time t includes:
Aircraft positions: $p_i(t) = [x_i(t), y_i(t)]$
Aircraft velocities: $v_i(t) = [v_{x_i}(t), v_{y_i}(t)]$
Thunderstorm cells information: $O^W(t)$
Remaining distance to exit waypoint: $d_t^{i_\text{Exit}}$
Thus, the state space is:

$$s_t^i = (p_i(t), v_i(t), O^W(t), d_t^{(i_\text{Exit})}) \tag{9}$$

4.1.2 Action Space. The action space for each aircraft is the heading change $\Delta h_i(t)$, which is a continuous variable within $[-30, 30]$ degrees:

$$a_t^i = \Delta h_i(t) \tag{10}$$

4.1.3 Reward Function. The reward function guides the learning process by evaluating the immediate benefit of actions. The key components are:

Separation assurance: $r_{i(t)}^{\text{sepa}}$ assigns a negative reward if the distance between aircraft i and any other aircraft j falls below the minimum separation distance $d_{\min}^{\text{flight}}$.

$$r_{i(t)}^{\text{sepa}} = \begin{cases} -1, & \text{if } \|p_i(t) - p_j(t)\| < d_{\min}^{\text{flight}}, \forall j \neq i \\ 0, & \text{otherwise} \end{cases} \tag{11}$$

Dynamic thunderstorm cell avoidance: $r_{i(t)}^{W}$ assigns a negative reward for loss of separation with a thunderstorm cell.

$$r_{i(t)}^{W} = \begin{cases} -1, & \text{if } \|p_i(t) - p_{O^W}(t)\| < d_{\min}^{W} + R_{\text{major_axis}}^{W}, \\ & \forall i \in \{1, \ldots, n\}, \forall t \in T \\ 0, & \text{otherwise} \end{cases} \tag{12}$$

Exit waypoint reachability: $r_{i(t)}^{\text{exit}}$ assigns a positive reward for reaching the designated exit waypoint.

$$r_{i(t)}^{\text{exit}} = \begin{cases} 1, & \text{if } p_i(t) = w_i \\ 0, & \text{otherwise} \end{cases} \tag{13}$$

Heading change discouragement: $r_{i(t)}^{\text{heading}}$ assigns a negative reward to minimize course changes. Here h^{interval} is the interval of heading change, which acts as a normalization term making the reward fall in the set $\{-1, 0\}$.

$$r_{i(t)}^{\text{heading}} = -\frac{\Delta h_i(t)}{h^{\text{interval}}} \tag{14}$$

Rerouting distance minimization: $r_{i(t)}^{\text{dist}}$ assigns a negative reward proportional to rerouting distance.

$$r_{i(t)}^{\text{dist}} = -\frac{\|p_i(t) - w_i\|}{\|p_i(t_0) - w_i\|} \tag{15}$$

where $p_i(t)$ is the current position of aircraft i, $p_i(t_0)$ is the position of aircraft i at the start of rerouting (when it enters the affected

airspace), and w_i is the position of its corresponding exit waypoint. The reward $r_{i(t)}^{\text{dist}}$ falls in the range of $[-1, 0]$.

The significance of each reward is weighted (e.g., separation assurance and storm avoidance given higher priority). The total system reward at time t is computed by summing the individual weighted rewards across all agents:

$$r_t^{\text{total}} = \sum_{i=1}^{n} \Big(\omega_{\text{sepa}} r_i^{\text{sepa}}(t) + \omega_W r_i^W(t) + \omega_{\text{exit}} r_i^{\text{exit}}(t) + \omega_{\text{heading}} r_i^{\text{heading}}(t) + \omega_{\text{dist}} r_i^{\text{dist}}(t) \Big) \tag{16}$$

Weights are assigned based on task priority, with separation assurance and storm avoidance receiving the highest importance, as $\omega_{\text{sepa}} = 10$ and $\omega_W = 8$. Weights of heading change and rerouting are given as 1 and 0.5, respectively. While reaching the exit waypoint is weighted as 10, which is the only positive reward to guide rerouted aircraft to rejoin their remaining flight route.

4.2 Independent Deep Deterministic Policy Gradient (IDDPG) Algorithm

Deep Deterministic Policy Gradient (DDPG) is well-suited for large, continuous action spaces, making it effective for air traffic management (ATM) tasks [22]. However, the decentralized Markov Decision Process (DEC-MDP) framework, which we model this problem, demands a multi-agent cooperative approach that standard DDPG, designed for single-agent environments, cannot fulfill. We therefore use the Independent Deep Deterministic Policy Gradient (IDDPG) framework for multi-agent cooperation [18].

The IDDPG algorithm enables each aircraft to learn an optimal policy using an actor-critic architecture. Each agent i has an actor network $\mu_i(s_t^i \mid \theta^{\mu_i})$ that maps its state s_t^i to a continuous action a_t^i, parameterized by θ^{μ_i}. Concurrently, each agent has a critic network $Q_i(s_t^i, a_t^i \mid \theta^{Q_i})$ that evaluates the quality of the chosen action a_t^i by estimating the expected return (Q-value), parameterized by θ^{Q_i}. The goal is to optimize policies and value functions to maximize expected cumulative reward $E\left[\sum_{t=0}^{T} \gamma^t r_{i(t)}\right]$, where γ is the discount factor. This optimization is performed through iterative updates of the policy and critic network parameters using gradient ascent and descent, and the policy gradient $\nabla_{\theta^{\mu_i}} J(\mu_i)$ and loss function $L(\theta^{Q_i})$ are calculated by:

$$\nabla_{\theta^{\mu_i}} J(\mu_i) = E_D \left[\nabla_{a_t^i} Q_i(s_t^i, a_t^i \mid \theta^{Q_i}) \nabla_{\theta^{\mu_i}} \mu_i(s_t^i \mid \theta^{\mu_i}) \right] \tag{17}$$

$$L(\theta^{Q_i}) = E_D \Big[\Big(r_{i(t)} + \gamma Q_i'(s_{t+1}^i, \mu_i'(s_{t+1}^i \mid \theta^{\mu_i'}) \mid \theta^{Q_i'}) - Q_i(s_t^i, a_t^i \mid \theta^{Q_i}) \Big)^2 \Big] \tag{18}$$

where D is the experience replay, μ_i' and Q_i' denote target actor and critic neural networks, which are parameterized by $\theta^{\mu_i'}$ and $\theta^{Q_i'}$.

Figure 2 illustrates the architecture and training process of the IDDPG algorithm. The training process begins with the Experience Replay, storing state transitions in a buffer to stabilize learning. Mini-batches sampled from the buffer are used to update the actor and critic networks. The IDDPG framework integrates shared actor and critic networks, allowing agents to learn from a consistent set of experiences. This approach mitigates non-stationarity, which occurs when agents independently update policies. By sharing neural networks and experiences, agents achieve more stable learning and policy convergence. Actor and critic optimizers are used to refine networks. The actor optimizer adjusts parameters to maximize expected rewards, while the critic optimizer refines Q-value estimation accuracy. Target networks are used to enhance stability, with parameters updated gradually via a soft update mechanism. This ensures stable references for calculating future values, leading to a robust training process.

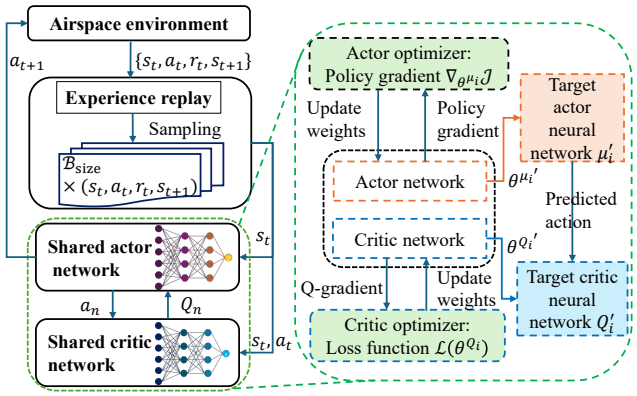


Figure 2: Training process of proposed IDDPG framework. The shared actor and critic networks enhance coordination and convergence efficiency, leading to a more coordinated decision-making process among multiple agents.

5 EXPERIMENTAL RESULTS

This section presents a comprehensive evaluation of our algorithm. We begin by describing the simulation environment configurations and hyperparameters used. We then analyze the training process of the IDDPG framework to determine optimal convergence settings, validate its performance in real-world scenarios, and assess generalization capabilities using standard air route structures [13].

5.1 Experimental Settings

The training scenarios simulated complex interactions involving airspace structures, aircraft, and dynamic thunderstorm cells within a 200 × 200 square nautical miles (nm) airspace. Five aircraft were assigned to routes, maintaining a constant speed of 400 knots with a minimum separation of 5 nm, while heading adjustments ranged from [-30, 30] degrees per time step. Two dynamic thunderstorm cells were introduced with varying speed, direction, and shape, emphasizing real-world adaptability. The environment parameters and hyperparameters are summarized in Table 1 and Table 2.

To evaluate our multi-agent coordination framework, we simulated a dynamic airspace environment with multiple aircraft and evolving thunderstorm cells. Aircraft enter the airspace at designated entry points at randomized intervals, each with a specific destination. Thunderstorm cells are introduced and evolve over time. Aircraft agents leverage decentralized learning to optimize

Actual airspace structutre can be non-symmetric. However, the authros have considered a symnmertic airspace structure (200x200)

their trajectories and maintain safe separation. The experiment terminates either when all aircraft reach their exit waypoints safely or when a conflict occurs.

Table 1: Simulation environment configurations.

Category	Parameter	Value/description
Airspace	Phase	En route
	Size of environment	200 nm × 200 nm
Aircraft	Speed	400 knots
	Altitude	Constant
	Heading change range	[-30, 30] degrees
	Separation minima	5 nm
Storm cell	Speed	[50, 90] knots
	Movement direction	[0, 359] degrees
	Size (radius)	[10, 25] nm
	Separation with aircraft	5 nm
	Update frequency	5 time steps
Duration	Time step	12 seconds
	Maximum steps	150
Hardware	CPU	Intel i9-11900
	GPU	NVIDIA GeForce RTX 3080
	Memory	32GB
Software	Custom-built simulator in Python 3.9	

Table 2: Hyperparameter settings for IDDPG algorithm.

Parameter	Value
Minibatch size	512
Replay buffer size	100000
Actor learning rate	0.0001
Critic learning rate	0.0001
Discount factor	0.95
Number of training episodes	20000
Soft update rate	0.01
Target network update frequency	1
Maximum play per episode	200
Noise level	$1 \longrightarrow 0.03$

Table 3: Performance Metrics.

Metric	Description
Aircraft LOS rate	$n_{\text{LOS aircraft}}/n_{\text{total aircraft}}$
Thunderstorm LOS rate	$n_{\text{LOS storm}}/n_{\text{total aircraft}}$
Goal reach rate	$n_{\text{all reach}}/n_{\text{total aircraft}}$
Flight distance ratio	Actual distance/planned route distance

Table 3 outlines performance metrics, evaluating safety and efficiency. Safety metrics include loss of separation (LOS) incidents, while efficiency metrics cover goal reach rates and flight distance ratios, indicating the effectiveness of planning and rerouting.

5.2 Model Training Results

We evaluated the effectiveness of different state observations within the IDDPG algorithm, as selecting optimal inputs is crucial for robust performance. Four types of observations were compared: (i) aircraft's own state, (ii) weather radar data, (iii) nearby aircraft's state, and (iv) ADS-B sensor data. We tested four input variations: baseline IDDPG (aircraft's own state and weather radar), IDDPG-n (including other aircraft's state), IDDPG-s (adding ADS-B data), and IDDPG-ns (all inputs combined).

The training curves (Figure 3) show that the baseline IDDPG had the slowest convergence and lowest rewards, indicating limited performance in complex scenarios. Adding other aircraft's state in IDDPG-n improved conflict avoidance ability, but performance plateaued under higher traffic density scenarios. IDDPG-s, incorporating sensor data, achieved the best results, demonstrating robustness and scalability under evolving conditions. The comprehensive IDDPG-ns model failed to outperform IDDPG-s, likely due to overfitting from added complexity.

These results highlight the importance of input selection in multi-agent systems for air traffic management, demonstrating that more data does not always lead to better outcomes. The IDDPG-s model effectively balances comprehensive input while maintaining efficiency, making it the optimal choice for real-time, multi-aircraft trajectory planning under diverse weather conditions. This work not only offers a scalable solution for ATM but also contributes to the autonomous multi-agent system community by showcasing the real-world potential of multi-agent reinforcement learning in dynamic and uncertain environments

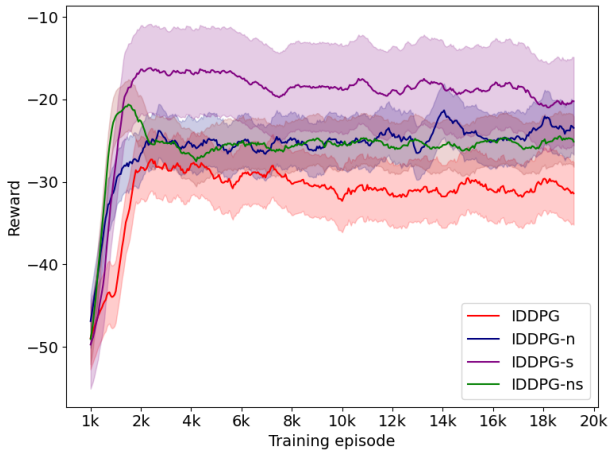


Figure 3: Training curves for different state variations of the IDDPG algorithm. The plot compares performance across different state configurations, demonstrating how the inclusion of key state features impacts learning stability and rewards.

5.3 Effectiveness and Scalability

We evaluate the IDDPG-s model in a real-world en route airspace, focusing on its scalability and effectiveness for multi-agent autonomous trajectory planning. The scenario, depicted in Figure 4,

is situated within an oceanic part of the flight information region (FIR), lacking radar coverage and ATC services, making our multi-agent approach particularly critical for high-density traffic under adverse weather events.

The selected 200 × 200 nm airspace includes three intersecting air routes (N884, M767, and M768) with major waypoints LAGOT and TODAM. Aircraft traffic enters and exits through designated waypoints with longitudinal separation reduced to 5 minutes to simulate a doubled capacity scenario, introducing four aircraft from three entry points. Two dynamic thunderstorm cells, with varying speed and size, move across the region from northwest to southeast.

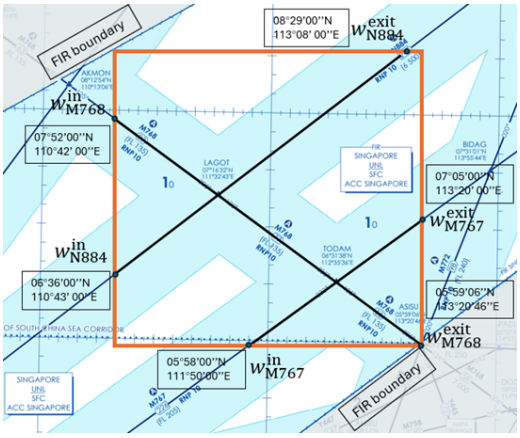


Figure 4: A real-world en route airspace area for experimental testing. The chosen airspace (within yellow box) includes three major airways intersecting at two critical waypoints. This area is situated within an oceanic part of the flight information region, lacking radar coverage and air traffic control service, making autonomous multi-agent approach particularly critical for coordinated decision making among multiple aircraft.

Figure 5 shows four conflict-free trajectories for aircraft a0, a1, a2, and a3 navigating around two dynamic thunderstorm cells. The aircraft initially follow predetermined routes, then adjust dynamically in response to the evolving positions of the thunderstorms. The final results indicate that all aircraft successfully navigate around thunderstorm cells while maintaining safe separation and reaching their exit waypoints safely. These results demonstrate the practical effectiveness of the proposed model in managing complex air traffic autonomously, showing potential for innovative ATM applications. Note that the enter and exit waypoints are represented by the symbols of solid triangle and star, respectively. And the dynamics of the thunderstorm cells are depicted using evolving contours, with decreasing transparency to represent their progression over different time steps.

To evaluate scalability, we conducted additional simulations with increasing numbers of aircraft (5, 6, 7, and 8) under the same dynamic weather conditions, with 100 runs for each scenario. Table 4 summarizes the results. The LOS rate between aircraft remained low, slightly increasing to 1% with 7 aircraft. However, LOS with thunderstorm cells increased to 5% with 8 aircraft, attributed to the

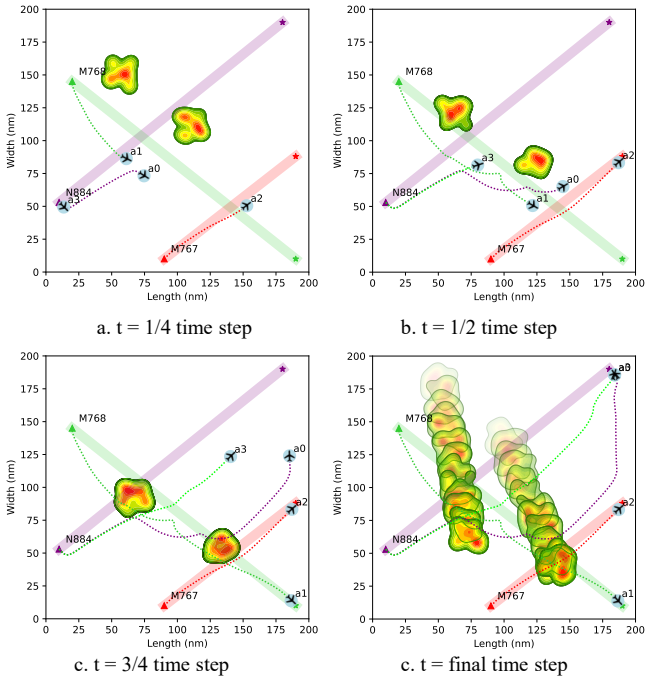


Figure 5: Conflict-free trajectories for multiple aircraft and thunderstorm cells at different time steps in the real-world airspace structure. Results show how each aircraft successfully avoids dynamic cells and maintains safe separation over time.

reward function prioritizing aircraft conflict avoidance over storm cell conflicts. The goal reach rate decreased from 100% to 95% with increased density, reflecting the increased difficulty in navigating safely as airspace becomes crowded. Flight distance ratios showed minor increases (mean and median shown in Figure 6), indicating additional maneuvers required to maintain safety, though the model managed rerouting efficiently.

Table 4: Scalability analysis with increased number of aircraft in 100 test runs.

Performance metrics	Number of aircraft				
	4	5	6	7	8
Aircraft LOS rate	0	0	0	1%	0
Thunderstorm LOS rate	0	2%	2%	3%	5%
Goal reach rate	100%	98%	98%	96%	95%
Flight distance ratio (Mean/S.D.)	1.08 /0.06	1.17 /0.15	1.17 /0.16	1.16 /0.17	1.17 /0.17

5.4 Generalization Ability Test

We evaluated the generalization ability of the trained IDDPG model across complex airspace configurations using two representative air route structures (ARS-1 and ARS-2) based on EuroControl guidelines [13]. Aircraft numbers varied from 4 to 8, with larger thunderstorm cells (25 nm radius) to assess the model's robustness. Results

Table 5: Generalization analysis of proposed IDDPG algorithm in diverse air route structures under 100 test runs.

Performance metrics (Average)	Air route structures					
	ARS-1			ARS-2		
Number of AC in each testing	4	6	8	4	6	8
Aircraft LOS rate	0	0	0	0	0	0
Thunderstorm LOS rate	0	1%	2%	0	0	1%
Goal reach rate	100%	99%	98%	100%	100%	99%
Flight distance ratio (Mean/S.D.)	1.23 /0.16	1.24 /0.18	1.24 /0.19	1.19 /0.12	1.21 1.15	1.24 /0.18

Table 6: Performance of the multi agent-based Fast Marching Tree (FMT) algorithm in ARS-2 scenarios under 100 test runs.

Number of aircraft	4			6	8
Avoidance threshold (nm)	20	30	40	30	30
Aircraft LOS rate	7.31%	6.06%	5.21%	11.59%	16.78%
Thunderstorm LOS rate	0.05%	0.02%	0.04%	0.05%	0.05%
Goal reach rate	100%	100%	100%	97%	93%
Flight distance ratio (Mean/S.D.)	1.13 /0.02	1.16 /0.03	1.26 /0.07	1.25 /0.14	1.28 1.20
Computation time* (s)	3.36	5.04	8.32	18.99	41.29

*Computation time measures the average time needed for trajectory planning at each time step.

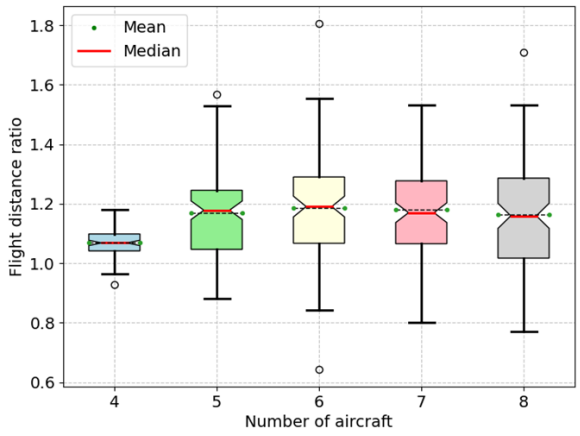


Figure 6: Flight distance ratios remained within an acceptable range with the increased number of aircraft agent.

(Table 5) show consistent zero LOS rates between aircraft across all scenarios, demonstrating the model's capability to maintain safety in dense traffic. The LOS rate with thunderstorm cells increased slightly, reaching 2% in ARS-1 and 1% in ARS-2 for 8 aircraft, due to the increased complexity of managing both traffic and storm cells simultaneously. Goal reach rates remained high, with only slight decreases as traffic density increased.

5.5 Algorithm Comparison Study

We compared the IDDPG algorithm to the latest multi agent-based Fast Marching Tree (FMT) algorithm [16] under increased number of aircraft agent (4, 6, and 8 aircraft) using the ARS-2 structure. Detailed performance metrics achieved by IDDPG and the FMT are provided in Table 5 and Table 6, respectively.

In 100 test runs, IDDPG achieved a 0% LOS rate between aircraft, whereas FMT's LOS rate increased to 16.78% for 8 aircraft (AC), highlighting IDDPG's superiority in maintaining safe separation under dynamic conditions. FMT's performance is sensitive to avoidance thresholds—smaller thresholds led to higher LOS rates but faster computation, while larger thresholds improved safety but increased time. In contrast, IDDPG dynamically adjusts actions, eliminating the need for fixed thresholds and offering better efficiency. As traffic density increased, FMT's computation time grew significantly, while IDDPG maintained efficient operation (with calculation time <20 ms). These results demonstrate the better performance of proposed algorithm in safety and scalability, making it well-suited for dynamic, time-sensitive airspace applications. This scalability and adaptability provide a strong case for its deployment in innovative multi-agent systems.

6 CONCLUSIONS AND FUTURE WORK

This study presents an innovative solution to the growing challenges of multi-aircraft cooperative trajectory planning in dynamic airspace impacted by climate change-induced thunderstorms. We developed a decentralized Markov Decision Process model and applied the Independent Deep Deterministic Policy Gradient (IDDPG) framework, enabling autonomous decision-making for multiple aircraft. Our design leverages multi-agent deep reinforcement learning to ensure robust conflict resolution, adaptability to evolving thunderstorm conditions, and efficient rerouting. The decentralized multi-agent approach addresses limitations of traditional centralized or single-agent methods, effectively managing real-time dynamic air traffic scenarios while maintaining low loss of separation rates and high safety standards. This contributes to both the ATM community by enhancing safety and operational efficiency, and the broader community by advancing autonomous multi-agent decision-making in safety-critical domains.

Future work will focus on expanding innovative applications in ATM and intelligent transportation, where such solutions are still very limited. Topics could range from enhancing robustness by incorporating uncertainties in storm evolution and trajectory prediction [17], expanding the model to different flight phases (e.g., terminal operations), and incorporating human-AI hybrid decision-making [31] to further improve applicability, thus, leading to a more safe and collaborative air traffic management solution.

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